

Power flow is:
 Battery -> Soft Start -> Cap bank -> MOSFETs -> Motor
 Battery -> +5V circuitry, +12V supply (They are placed before the soft-start)

5V circuit includes 5V isolated DCDC, Arduino, optocoupler, current sensor.
 12V circuit includes 12V DCDC, MOSFET drivers, Half-Bridge Controller, optocoupler (non-isolated)
 There is also a non-isolated +5V for the isolated voltage sensor.
 VDC is the connection from the output of the soft-start and filtering caps.

The 5V and 12V supply is connected before the soft-start, so the Arduino can control it.

Version 2 Edits:

- Added soft-start opto-isolator input resistor
- Replaced VOHXXX opto-isolators with TPLXXX with have a totem-pole output.
- Update creation date

Sheets contain subcircuits that interact with the overall system but are too complex to have on the one sheet and are functionally similar

Sheet: Filtering_Caps
 File: Filtering_Caps.sch

Sheet: DCDC_Converters
 File: DCDC_Converters.sch

Sheet: Misc
 File: Misc.sch

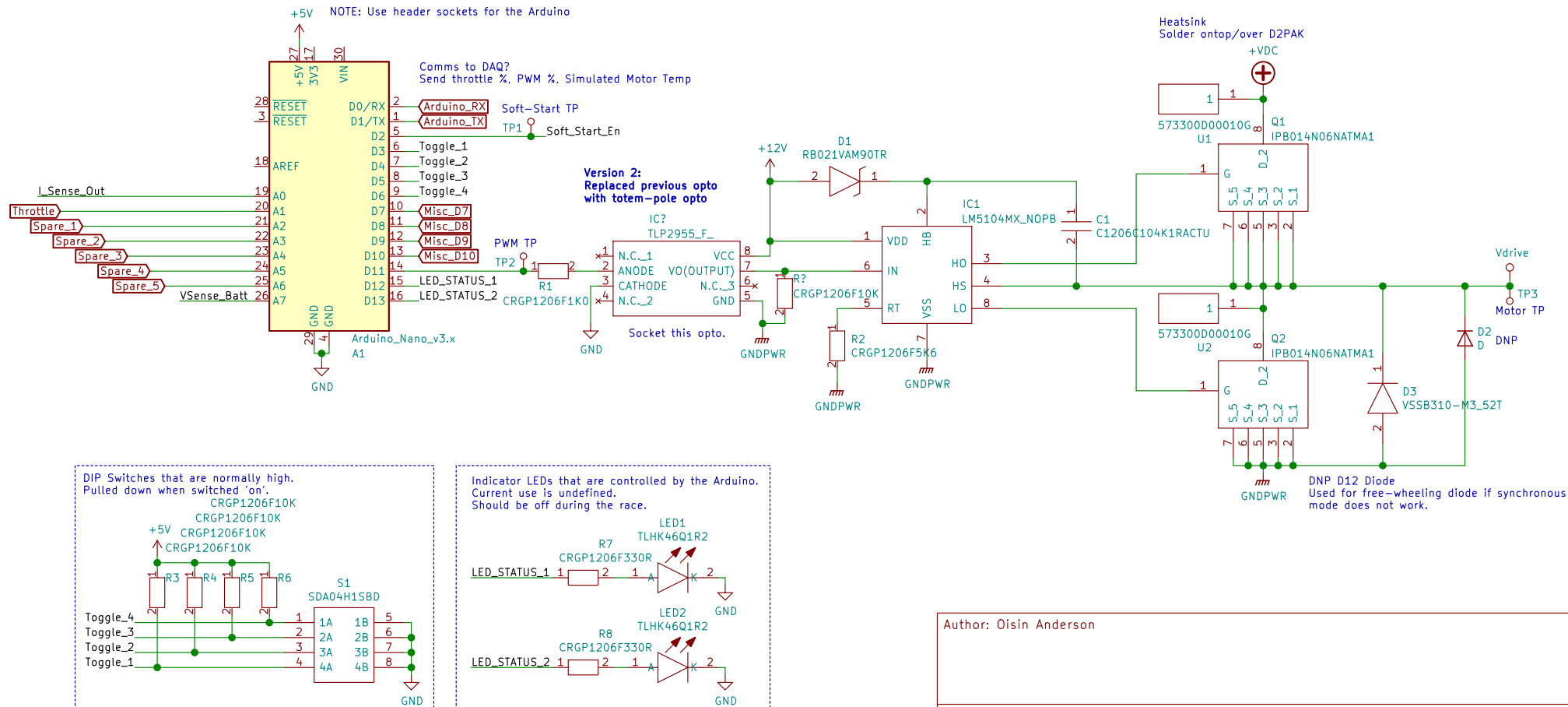
Sheet: JST_Conns
 File: JST_Conns.sch

These three sheets contain the current sensor (placed before the filtering caps), the soft-start circuit that is controlled by the Arduino, and the voltage sensor connected to the battery input

Sheet: Voltage-Sensor
 VOUT_Batt_Sense
 File: Voltage-Sensor.sch

Sheet: Current-Sensor
 PWR_IN PWR_OUT
 VOUT L_Sense_Out Soft_Start_En
 File: Current-Sensor.sch

Sheet: Soft-Start
 Pre-FET Post-FET
 MOSFET_EN
 File: Soft-Start.sch



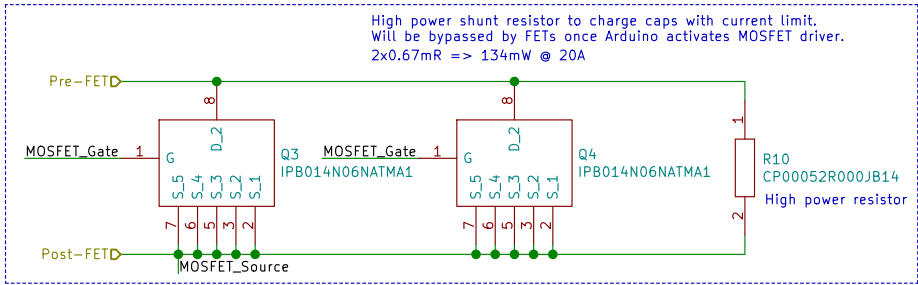
Author: Oisin Anderson

Sheet: /
 File: SMD_Sync_Buck_Controller.sch

Title: Geec Power Electronics: Synchronous Buck Converter

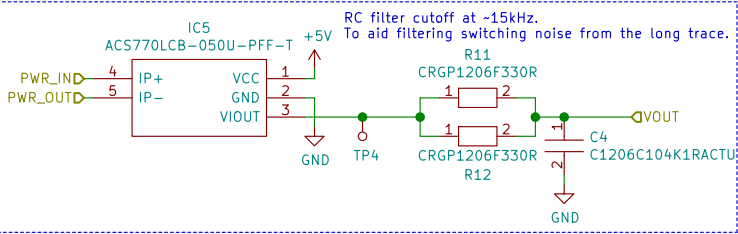
Size: A4
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Date:
 Rev:
 Id: 1/8

[illegible]

Rev: 1.0
Id: 2/8

Originally there was more circuitry for the current sensor, so it was on it's own page.
But it was replaced with a simpler one.



Sheet: /Current-Sensor/
File: Current-Sensor.sch

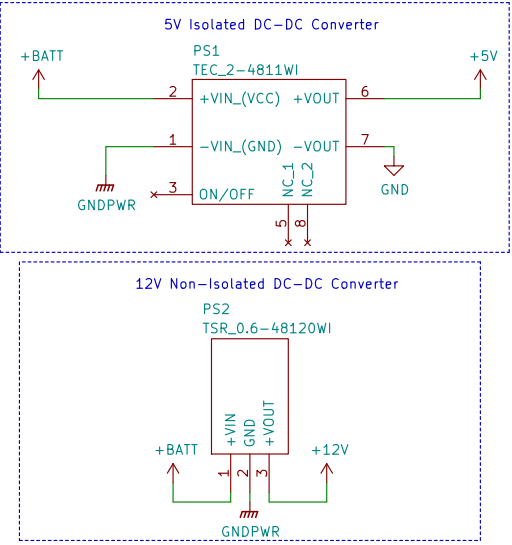
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Rev:
Id: 3/8

DC-DC Converters provide the power to the ICs around the board.
5V Goes to the Arduino, Current Sensor, Opto-isolator
12V Goes to the MOSFET driver, Half-Bridge Driver, Opto-isolator



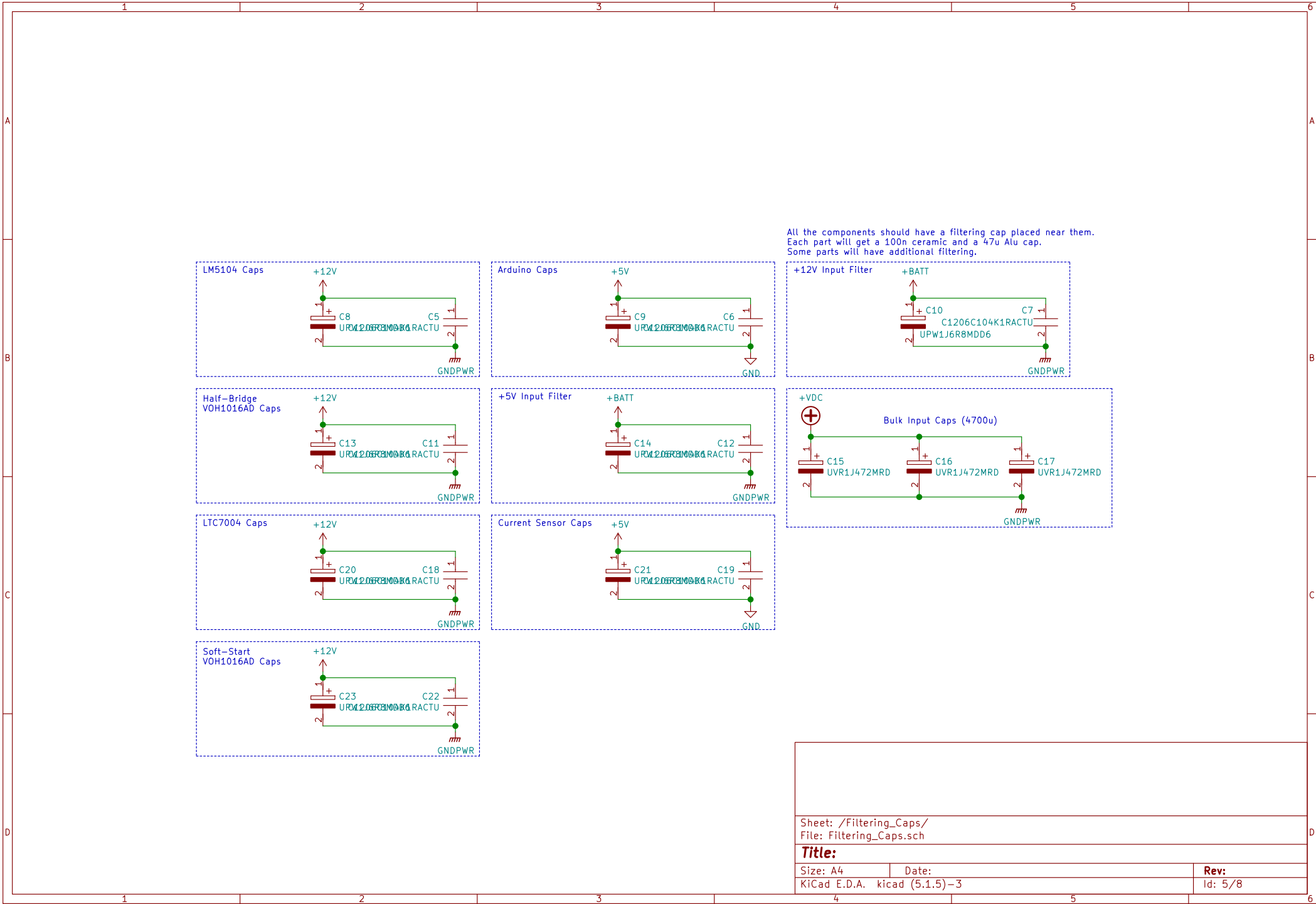
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File: DCDC_Converters.sch

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Date:

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Id: 4/8

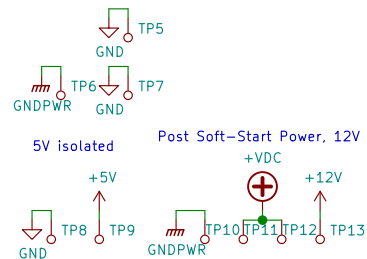


M4 mounting holes.
To be used for stand-offs.
To be placed on the perimeter of the board.

- H1
MountingHole
- H2
MountingHole
- H3
MountingHole
- H4
MountingHole

This sheet contains the parts that didn't really fit anywhere.

Ground TP placed around components
for easier debug/probing

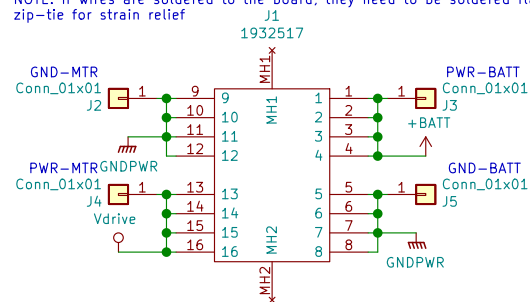


Geec logo to go on the front of the PCB

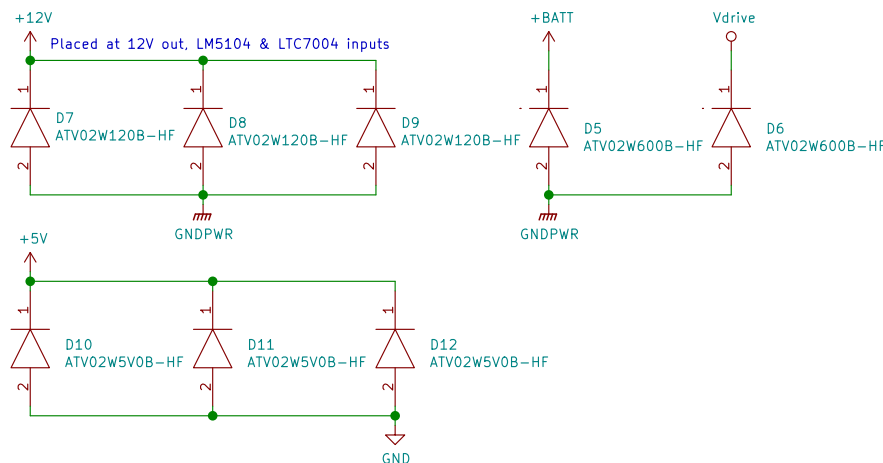


Main connector for power.
The 1932517 is a screw terminal. Does not need to be populated
if the wires are soldered to the board.

The J connectors are 10x5mm copper pads to solder the wires to.
NOTE: If wires are soldered to the board, they need to be soldered flat and with a
zip-tie for strain relief



TVS diodes.
5V, 12V, and 60V.
Place near sensitive components.



Sheet: /Misc/
File: Misc.sch

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Date:

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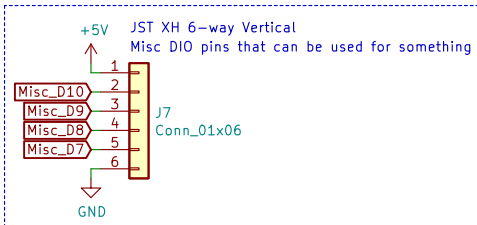
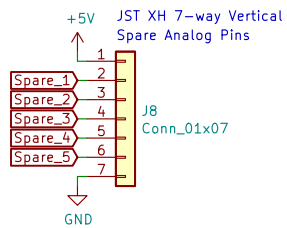
Rev:

Id: 6/8

The throttle is the main connector in use, the others are extra to use the unconnected Arduino pins.

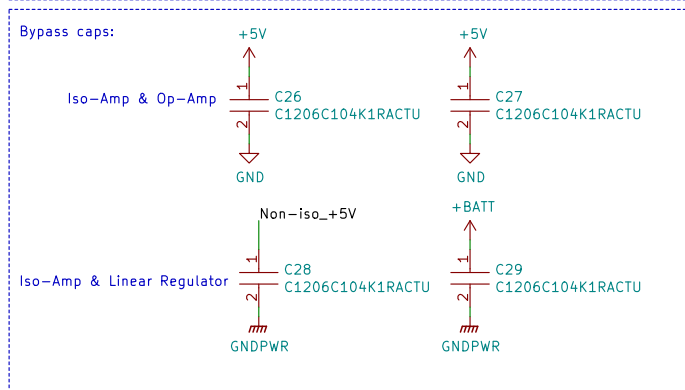
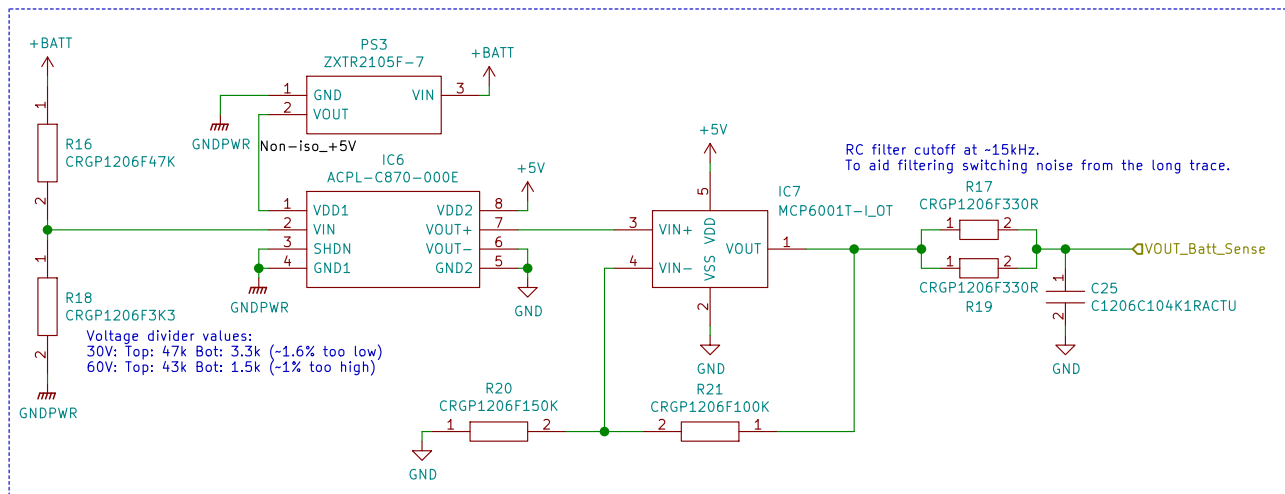
RC filter cutoff at ~15kHz.
To aid filtering switching noise from the long trace.

+5V
↑



Rev:
Id: 7/8

We only need a voltage sensor connected to the battery input, since we can calculate the voltage and current in the motor from the input power, voltage and duty cycle.
Op-Amp to amplify a 2V signal to a 5V signal to make use of the Arduino's full range ADC.
Linear regulator is used to power the battery side of the iso-amp because:
1: I physically have no room,
2: the power is low enough that it shouldn't contribute much to the losses.



Sheet: /Voltage-Sensor/
File: Voltage-Sensor.sch

Title:

Size: A4
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Rev:
Id: 8/8